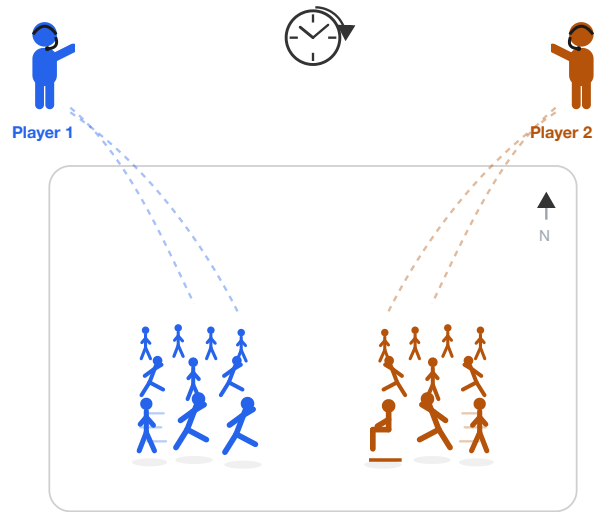


World Commander

Natural-language command of agent crowds in real-time strategy games, under latency and memory budgets

This proposal grew from Yubo Huang's interest and insight, under the guidance of Dr. Enmao Diao.



What we build: real-time, language-commanded agent crowds (here, two players commanding their own).

The long-term goal: the next revolutionary video game, played by voice from the commander's chair.

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1 Related Work: Efficiency, Untested at Game Speed

- The interface shipped once (Tom Clancy's EndWar (2008)), but on a 70-word grammar. Language models lifted that limit; they still **cannot act at game pace**.
- LLMs already play StarCraft II, by text (TextStarCraft II (NeurIPS 2024)) or screenshots (AVA (2026)), but only by **pausing or slowing the clock**.
- Real-time is only now being measured (VideoGameBench (2025): models collapse once the clock keeps running), and only for single agents, **not the efficiency methods**.
- Eviction, quantization, and distillation are tuned on perplexity, where they already drop the wrong instruction (Pitfalls of KV Cache Compression (ACL 2026)); none is **scored by win rate in a live RTS**, where that lost context loses the match.

| **How much does command cost, in latency and memory, at game speed, and how do we drive it down?**

2 Preliminaries

The background this work builds on:

Area	What it gives us
Reinforcement learning	A game is a Markov decision process: an agent acts, and win rate is the reward. We need to <i>understand</i> this to read the field; the work runs models by prompting, training none of its own.
Efficient LLM inference	KV-cache eviction (H2O, SnapKV, StreamingLLM, <u>OBCache</u>), structured pruning, quantization, distillation: the methods this work puts to the test.
Vision-language-action models	<u>π_0</u> : a slow vision-language backbone driving a fast action expert, trained by imitation. A model for splitting a slow strategic brain from fast executors.
The StarCraft II agent stack	<u>PySC2</u> , <u>TextStarCraft II</u> , LLM-PySC2: the environment we inherit rather than rebuild.
Tokenization beyond text	Byte-pair encoding and <u>graph tokenization</u> (ICLR 2026): the basis for a learned game-state tokenizer.

3 The Roadmap: Three Phases of One Program

World Commander is one **program**: an LLM executing a human commander's intent at game speed, under latency and memory budgets, matured in three **phases** over environments of growing complexity (a toy room → StarCraft II → multiplayer → a full game), under one shared **harness**. The end-state game is the long-term goal, not a deliverable; each phase answers a question its community already cares about, beyond the game.

Phase	Goal	Environment	Deliverable (one paper)
1 Benchmarks	Measure the cost of command at game speed	command arena → real-time SC2	which efficiency methods survive a live, closed-loop clock (eviction scored by win rate, not perplexity)
2 Methods	Drive that cost down	SC2 (harder) → multiplayer	the game-state tokenizer (fewer tokens per decision, hence less latency and memory); better eviction; action-level speculation
3 Interface	Ship a playable system	multiplayer → full game	voice-commanded play; crowd motion under budget (embodiment), deferred

4 The Command Arena

The first step: a warm-up task that validates the streaming-command infrastructure cheaply, before StarCraft II.

- Colour-tagged agents in a room; each moves in **one of four directions** on command.
- One command is trivial; the test is the **stream**: many, fast.
- Harder with **rate**, **compositional orders** ("everyone but the yellow one"), and **memory** ("the one I moved west").
- Measured: **grounding accuracy**, **command-to-action latency**, **deadline misses**.

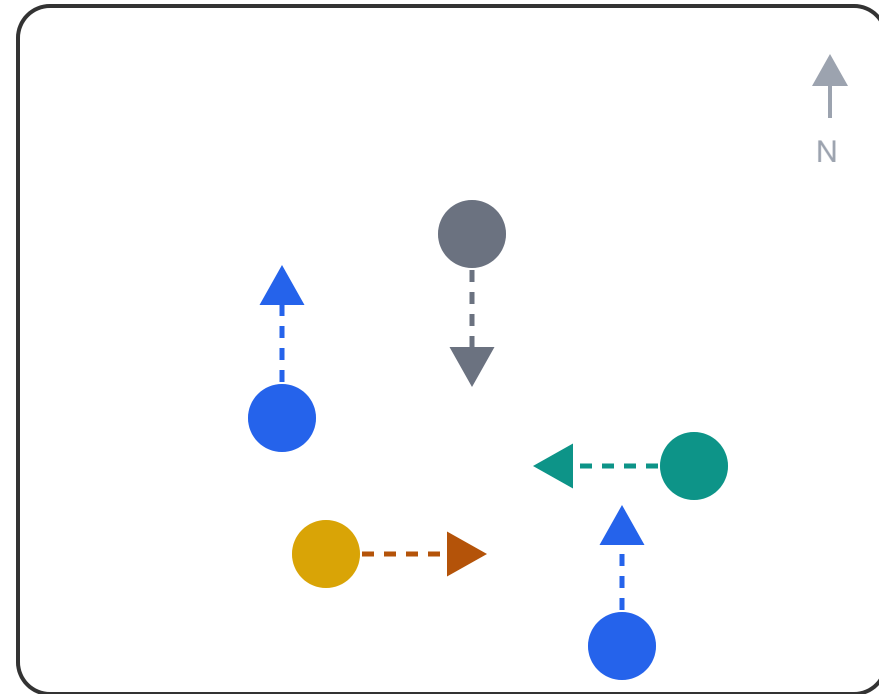


Figure 2: the command arena. Colour-tagged agents, each moving in one of four directions on command.

Appendix: Related-Work Landscape (1 of 2)

Agents at game speed, and the methods under test. The fuller field behind slides 1 and 2; the canonical review with every link is [LITERATURE.md](#).

Agents in real-time games. AlphaStar (Nature 2019) mastered StarCraft II by full reinforcement learning (no language, datacentre scale, fully autonomous); the precedent we deliberately do not repeat. LLMs now issue SC2 commands (TextStarCraft II, LLM-PySC2, AVA), but only with the clock paused or slowed. Real-time is the new axis: VideoGameBench and AgileThinker show models collapse once it keeps running. Closest to us, HLA, Adaptive Command, and DPT-Agent put a human in command with a fast-slow split, but cooperatively, or by steering a hand-built behaviour tree. CivRealm is the turn-based-strategy contrast.

Efficient inference, under latency and memory. The methods we put on the clock: KV-cache eviction (H2O, SnapKV, StreamingLLM, OBCache), plus action-level Speculative Actions, which cuts latency losslessly. The warning shots: Pitfalls of KV Cache Compression, DefensiveKV, and SideQuest show eviction silently drops the wrong token, and that its quality is task-dependent. WorldMemArena pushes agent-memory evaluation into a closed interaction loop: a different "memory" from the KV-cache, but the same methodological turn we make.

Appendix: Related-Work Landscape (2 of 2)

Fast-slow execution, game-state representation, and the Phase-3 motion stack.

Hierarchical and VLA fast-slow execution. π_0 pairs a slow vision-language brain with a fast action expert (up to 50 Hz); Fast-in-Slow folds both into one backbone; OpenVLA is the open "tokens-as-actions" baseline. This is the split we adapt for commander plus executors. Language-conditioned precedents: NL-to-StarCraft II grounding (2019) and CALVIN.

Game-state representation and world models. Toward fewer tokens per decision: Diao's graph tokenization (ICLR 2026), VQ-VAE, and IRIS / Δ -IRIS (context-aware delta tokenization). Action-conditioned world models, WHAM / WHAMM (Nature 2025), GameNGen, and DreamerV3, are the engines behind any what-if forecaster.

Crowd and real-time motion (Phase 3, embodiment). CrowdMoGen has an LLM plan a crowd's motion, but makes no real-time claim. Per-character real-time generators under budget: MotionLCM, MotionPCM, MotionStreamer, and MotionBricks (NVIDIA, SIGGRAPH 2026; 350k skills from one backbone). Single-character today; language-commanded crowds under one GPU is the gap we name.